

Finite samples have zero-dimensional Zariski closure

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Conjecture 2839 claims that the Zariski closure of a generic n -point sample in ambient dimension d has dimension $\min(n - 1, d)$. That formula fails already for $n = 2, d = 1$, for *every* two-point sample, so choosing a generic sample cannot repair it.

Theorem 1. *For any two distinct points a, b on the affine line over an algebraically closed field, their Zariski closure has dimension zero.*

Proof. The polynomial $(X - a)(X - b)$ has zero set exactly $\{a, b\}$, because the field has no zero divisors. Thus the sample itself is Zariski closed and is its own Zariski closure. Each singleton is also closed, being the zero set of $X - a$ or $X - b$.

A nonempty irreducible closed subset of $\{a, b\}$ must be contained in one of those singletons: apply irreducibility to the cover $\{a, b\} = \{a\} \cup \{b\}$. Hence the only nonempty irreducible closed subsets are the two singletons. There is a chain of length zero and no strict chain of length one. By the standard dimension definition using chains of nonempty irreducible closed subsets, the dimension is exactly zero. The asserted formula instead gives $\min(2 - 1, 1) = 1$. $\square$

The proof is uniform in a, b , including points in every nonempty generic locus of the configuration space. It does not substitute a specially chosen rational sample for a generic one. The affine span of two distinct points in the affine line does have dimension one, but affine span is a different operation from the Zariski closure named in both language versions of the source. No finite restriction on polynomial degree is part of the ordinary Zariski closure.

Formal verification. `DomainData` lists the ordinary integral-domain laws, all valid over any field; the main theorem is universally quantified in those laws. An integer model separately checks consistency without restricting that universal theorem. `Polynomial` contains precisely constant, variable, addition, multiplication and negation expressions. Every usual coefficient-list polynomial is embedded by Horner's construction, with a proved evaluation bridge.

Zariski closed sets are zero sets of arbitrary collections of these polynomials. Closure is the actual common-zero set $V(I(S))$ of all polynomials vanishing on the sample. The file proves inclusion, closedness, and the least-closed-superset property, then identifies the two-point closure exactly. Irreducibility quantifies over arbitrary closed covers. Dimension at least k is witnessed by a length- k strict chain of nonempty irreducible closed subsets. The proof classifies all such subsets inside the two-point set and excludes every chain of length one, while constructing one of length zero.

The final theorem rules out *any* distinct pair with the dimension predicted at $n = 2, d = 1$. It therefore excludes any putative generic instance as well. Lean 4.19.0 verifies the complete polynomial, closure, irreducibility, and dimension chain with only `Std`, no custom axioms, no admitted declarations, and no native evaluation proof.